

Result (cost=74290.44..74290.45 rows=1 width=64) (actual time=744.462..749.461 rows=1.00 loops=1)
 Buffers: shared hit=37764
 InitPlan 2
 -> Limit (cost=37086.19..37145.22 rows=1 width=32) (actual time=395.789..396.428 rows=1.00 loops=1)
 Buffers: shared hit=18882
 InitPlan 1
 -> Aggregate (cost=18695.52..18695.53 rows=1 width=8) (actual time=197.364..197.705 rows=1.00 loops=1)
 Buffers: shared hit=9441
 -> Finalize GroupAggregate (cost=18390.66..18683.38 rows=971 width=14) (actual time=197.205..197.692 rows=99.00 loops=1)
 Group Key: m.song_title
 Buffers: shared hit=9441
 -> Gather Merge (cost=18390.66..18662.02 rows=2330 width=14) (actual time=197.196..197.618 rows=297.00 loops=1)
 Workers Planned: 2
 Workers Launched: 2
 Buffers: shared hit=9441
 -> Sort (cost=17390.63..17393.06 rows=971 width=14) (actual time=190.975..190.988 rows=99.00 loops=3)
 Sort Key: m.song_title
 Sort Method: quicksort Memory: 28kB
 Buffers: shared hit=9441
 Worker 0: Sort Method: quicksort Memory: 28kB
 Worker 1: Sort Method: quicksort Memory: 28kB
 -> Partial HashAggregate (cost=17332.74..17342.45 rows=971 width=14) (actual time=190.877..190.902 rows=99.00 loops=3)
 Group Key: m.song_title
 Batches: 1 Memory Usage: 65kB
 Buffers: shared hit=9425
 Worker 0: Batches: 1 Memory Usage: 65kB
 Worker 1: Batches: 1 Memory Usage: 65kB
 -> Hash Join (cost=2905.03..16608.74 rows=144801 width=6) (actual time=15.665..164.425 rows=116176.00 loops=3)
 Hash Cond: (p.song_id = m.song_id)
 Buffers: shared hit=9425
 -> Parallel Hash Join (cost=2433.03..15756.48 rows=144801 width=4) (actual time=10.122..131.614 rows=116176.00 loops=3)
 Hash Cond: (p.user_id = u.user_id)
 Buffers: shared hit=8684
 -> Parallel Seq Scan on

playlists p (cost=0.00..12561.33 rows=290317 width=20) (actual time=0.020..44.902 rows=232246.33 loops=3)

Filter: (added_date >

'2023-01-01'::date)

Rows Removed by

Filter: 101087

Buffers: shared hit=7353

-> Parallel Hash

(cost=2066.29..2066.29 rows=29339 width=16) (actual time=9.929..9.930 rows=16656.67 loops=3)

Buckets: 65536

Batches: 1 Memory Usage: 2880kB

Buffers: shared hit=1331

-> Parallel Seq Scan

on users u (cost=0.00..2066.29 rows=29339 width=16) (actual time=0.013..5.817 rows=16656.67 loops=3)

Filter:

((subscription_type)::text = 'premium'::text)

Rows Removed by

Filter: 16677

Buffers: shared

hit=1331

-> Hash (cost=347.00..347.00

rows=10000 width=10) (actual time=5.394..5.395 rows=10000.00 loops=3)

Buckets: 16384 Batches: 1

Memory Usage: 560kB

Buffers: shared hit=741

-> Seq Scan on music m

(cost=0.00..347.00 rows=10000 width=10) (actual time=0.092..2.776 rows=10000.00 loops=3)

Buffers: shared hit=741

-> Subquery Scan on sub2 (cost=18390.66..18685.82 rows=5 width=32) (actual time=395.787..396.085 rows=1.00 loops=1)

Buffers: shared hit=12940

-> Finalize GroupAggregate (cost=18390.66..18685.81 rows=5 width=14) (actual time=395.779..396.076 rows=1.00 loops=1)

Group Key: m_1.song_title

Filter: (count(*) = (InitPlan 1).col1)

Rows Removed by Filter: 2

Buffers: shared hit=12940

-> Gather Merge (cost=18390.66..18662.02 rows=2330 width=14) (actual time=198.398..198.698 rows=10.00 loops=1)

Workers Planned: 2

Workers Launched: 2

Buffers: shared hit=9441

-> Sort (cost=17390.63..17393.06 rows=971 width=14) (actual time=191.430..191.439 rows=67.33 loops=3)

Sort Key: m_1.song_title

Sort Method: quicksort Memory: 28kB

Buffers: shared hit=9441

Worker 0: Sort Method: quicksort Memory: 28kB
 Worker 1: Sort Method: quicksort Memory: 28kB
 -> Partial HashAggregate
 (cost=17332.74..17342.45 rows=971 width=14) (actual time=191.332..191.355
 rows=99.00 loops=3)
 Group Key: m_1.song_title
 Batches: 1 Memory Usage: 65kB
 Buffers: shared hit=9425
 Worker 0: Batches: 1 Memory Usage:
 65kB
 Worker 1: Batches: 1 Memory Usage:
 65kB
 -> Hash Join (cost=2905.03..16608.74
 rows=144801 width=6) (actual time=16.529..164.697 rows=116176.00 loops=3)
 Hash Cond: (p_1.song_id =
 m_1.song_id)
 Buffers: shared hit=9425
 -> Parallel Hash Join
 (cost=2433.03..15756.48 rows=144801 width=4) (actual time=11.021..132.112
 rows=116176.00 loops=3)
 Hash Cond: (p_1.user_id =
 u_1.user_id)
 Buffers: shared hit=8684
 -> Parallel Seq Scan on
 playlists p_1 (cost=0.00..12561.33 rows=290317 width=20) (actual
 time=0.013..45.958 rows=232246.33 loops=3)
 Filter: (added_date >
 '2023-01-01'::date)
 Rows Removed by Filter:
 101087
 Buffers: shared hit=7353
 -> Parallel Hash
 (cost=2066.29..2066.29 rows=29339 width=16) (actual time=10.811..10.812
 rows=16656.67 loops=3)
 Buckets: 65536 Batches:
 1 Memory Usage: 2880kB
 Buffers: shared hit=1331
 -> Parallel Seq Scan on
 users u_1 (cost=0.00..2066.29 rows=29339 width=16) (actual time=0.017..6.419
 rows=16656.67 loops=3)
 Filter:
 ((subscription_type)::text = 'premium'::text)
 Rows Removed by
 Filter: 16677
 Buffers: shared
 hit=1331
 -> Hash (cost=347.00..347.00
 rows=10000 width=10) (actual time=5.431..5.431 rows=10000.00 loops=3)
 Buckets: 16384 Batches: 1
 Memory Usage: 560kB

Buffers: shared hit=741
-> Seq Scan on music m_1
(cost=0.00..347.00 rows=10000 width=10) (actual time=0.083..2.736 rows=10000.00 loops=3)

Buffers: shared hit=741

InitPlan 4

-> Limit (cost=37086.19..37145.22 rows=1 width=32) (actual time=348.667..353.025 rows=1.00 loops=1)

Buffers: shared hit=18882

InitPlan 3

-> Aggregate (cost=18695.52..18695.53 rows=1 width=8) (actual time=171.002..175.038 rows=1.00 loops=1)

Buffers: shared hit=9441

-> Finalize GroupAggregate (cost=18390.66..18683.38 rows=971 width=14) (actual time=170.856..175.025 rows=99.00 loops=1)

Group Key: m_2.song_title

Buffers: shared hit=9441

-> Gather Merge (cost=18390.66..18662.02 rows=2330 width=14) (actual time=170.847..174.955 rows=297.00 loops=1)

Workers Planned: 2

Workers Launched: 2

Buffers: shared hit=9441

-> Sort (cost=17390.63..17393.06 rows=971 width=14) (actual time=165.729..165.740 rows=99.00 loops=3)

Sort Key: m_2.song_title

Sort Method: quicksort Memory: 28kB

Buffers: shared hit=9441

Worker 0: Sort Method: quicksort Memory:

28kB

Worker 1: Sort Method: quicksort Memory:

28kB

-> Partial HashAggregate

(cost=17332.74..17342.45 rows=971 width=14) (actual time=165.629..165.658 rows=99.00 loops=3)

Group Key: m_2.song_title

Batches: 1 Memory Usage: 65kB

Buffers: shared hit=9425

Worker 0: Batches: 1 Memory Usage:

65kB

Worker 1: Batches: 1 Memory Usage:

65kB

-> Hash Join (cost=2905.03..16608.74 rows=144801 width=6) (actual time=14.442..142.174 rows=116176.00 loops=3)

Hash Cond: (p_2.song_id =

m_2.song_id)

Buffers: shared hit=9425

-> Parallel Hash Join

(cost=2433.03..15756.48 rows=144801 width=4) (actual time=9.815..113.855 rows=116176.00 loops=3)

Hash Cond: (p_2.user_id =

u_2.user_id)

Buffers: shared hit=8684

-> Parallel Seq Scan on

playlists p_2 (cost=0.00..12561.33 rows=290317 width=20) (actual
time=0.012..39.771 rows=232246.33 loops=3)

Filter: (added_date >

'2023-01-01'::date)

Rows Removed by

Filter: 101087

Buffers: shared hit=7353

-> Parallel Hash

(cost=2066.29..2066.29 rows=29339 width=16) (actual time=9.628..9.628
rows=16656.67 loops=3)

Buckets: 65536

Batches: 1 Memory Usage: 2912kB

Buffers: shared hit=1331

-> Parallel Seq Scan

on users u_2 (cost=0.00..2066.29 rows=29339 width=16) (actual time=0.014..5.528
rows=16656.67 loops=3)

Filter:

((subscription_type)::text = 'premium'::text)

Rows Removed by

Filter: 16677

Buffers: shared

hit=1331

-> Hash (cost=347.00..347.00

rows=10000 width=10) (actual time=4.531..4.531 rows=10000.00 loops=3)

Buckets: 16384 Batches: 1

Memory Usage: 560kB

Buffers: shared hit=741

-> Seq Scan on music m_2

(cost=0.00..347.00 rows=10000 width=10) (actual time=0.098..2.359 rows=10000.00
loops=3)

Buffers: shared hit=741

-> Subquery Scan on sub2_1 (cost=18390.66..18685.82 rows=5
width=32) (actual time=348.666..348.987 rows=1.00 loops=1)

Buffers: shared hit=13018

-> Finalize GroupAggregate (cost=18390.66..18685.81 rows=5
width=14) (actual time=348.660..348.980 rows=1.00 loops=1)

Group Key: m_3.song_title

Filter: (count(*) = (InitPlan 3).col1)

Rows Removed by Filter: 98

Buffers: shared hit=13018

-> Gather Merge (cost=18390.66..18662.02 rows=2330
width=14) (actual time=177.487..177.889 rows=297.00 loops=1)

Workers Planned: 2

Workers Launched: 2

Buffers: shared hit=9441

-> Sort (cost=17390.63..17393.06 rows=971
width=14) (actual time=172.099..172.110 rows=99.00 loops=3)

Sort Key: m_3.song_title
 Sort Method: quicksort Memory: 28kB
 Buffers: shared hit=9441
 Worker 0: Sort Method: quicksort Memory: 28kB
 Worker 1: Sort Method: quicksort Memory: 28kB
 -> Partial HashAggregate
 (cost=17332.74..17342.45 rows=971 width=14) (actual time=171.918..171.940
 rows=99.00 loops=3)

 Group Key: m_3.song_title
 Batches: 1 Memory Usage: 65kB
 Buffers: shared hit=9425
 Worker 0: Batches: 1 Memory Usage:
 65kB
 Worker 1: Batches: 1 Memory Usage:
 65kB
 -> Hash Join (cost=2905.03..16608.74
 rows=144801 width=6) (actual time=15.308..147.781 rows=116176.00 loops=3)
 Hash Cond: (p_3.song_id =
 m_3.song_id)
 Buffers: shared hit=9425
 -> Parallel Hash Join
 (cost=2433.03..15756.48 rows=144801 width=4) (actual time=10.452..118.413
 rows=116176.00 loops=3)
 Hash Cond: (p_3.user_id =
 u_3.user_id)
 Buffers: shared hit=8684
 -> Parallel Seq Scan on
 playlists p_3 (cost=0.00..12561.33 rows=290317 width=20) (actual
 time=0.013..40.973 rows=232246.33 loops=3)
 Filter: (added_date >
 '2023-01-01'::date)
 Rows Removed by Filter:
 101087
 Buffers: shared hit=7353
 -> Parallel Hash
 (cost=2066.29..2066.29 rows=29339 width=16) (actual time=10.229..10.230
 rows=16656.67 loops=3)
 Buckets: 65536 Batches:
 1 Memory Usage: 2912kB
 Buffers: shared hit=1331
 -> Parallel Seq Scan on
 users u_3 (cost=0.00..2066.29 rows=29339 width=16) (actual time=0.014..5.881
 rows=16656.67 loops=3)
 Filter:
 ((subscription_type)::text = 'premium'::text)
 Rows Removed by
 Filter: 16677
 Buffers: shared
 hit=1331
 -> Hash (cost=347.00..347.00

rows=10000 width=10) (actual time=4.742..4.743 rows=10000.00 loops=3)

Buckets: 16384 Batches: 1

Memory Usage: 560kB

Buffers: shared hit=741

-> Seq Scan on music m_3

(cost=0.00..347.00 rows=10000 width=10) (actual time=0.111..2.447 rows=10000.00 loops=3)

Buffers: shared hit=741

Planning:

Buffers: shared hit=56

Planning Time: 1.562 ms

Execution Time: 750.217 ms